

Question. (Erdős #125) Let $A = \{\sum \epsilon_k 3^k : \epsilon_k \in \{0, 1\}\}$ be the set of integers which have only the digits 0,1 when written base 3, and $B = \{\sum \epsilon_k 4^k : \epsilon_k \in \{0, 1\}\}$ be the set of integers which have only the digits 0,1 when written base 4. Does $A + B$ have positive lower density?

Proof. We show that the answer is no, the lower density is zero. Let A and B be defined as the sets of integers whose base 3 and base 4 representations, respectively, contain only the digits 0 and 1. We will show that the lower density of $A + B$ is zero.

For any integer $a \in A$, we can decompose it at the k -th digit into a top and bottom part, writing $a = 3^k a_{top} + a_{bot}$. Because a uses only the digits 0 and 1, both a_{top} and a_{bot} must also belong to A . Furthermore, the maximum possible value for a_{bot} is the integer consisting of k ones in base 3, which provides the strict upper bound $a_{bot} \leq (3^k - 1)/2$. By applying the exact same logic to any $b \in B$ at scale m , we can write $b = 4^m b_{top} + b_{bot}$, yielding the analogous bound $b_{bot} \leq (4^m - 1)/3$.

To evaluate the density of $A + B$ up to a large scale, we define a threshold $M \cdot N_0$, where $M = \min(3^k, 4^m)$ for some integers k and m . Consider any element $x \in A + B$ such that $x < M \cdot N_0$. By definition, $x = a + b$ for some $a \in A$ and $b \in B$. Using the previously established decompositions, we can express this sum as $x = 3^k a_{top} + a_{bot} + 4^m b_{top} + b_{bot}$. Assume without loss of generality that $3^k \leq 4^m$, so $M = 3^k$. We can rewrite this expression to factor out M , yielding $x = 3^k(a_{top} + b_{top}) + c$, where the remainder term is $c = (4^m - 3^k)b_{top} + a_{bot} + b_{bot}$.

Let $y = a_{top} + b_{top}$. Since $a_{top} \in A$ and $b_{top} \in B$, it naturally follows that $y \in A + B$. Furthermore, because $x < 3^k N_0$, we must have $y < N_0$, which in turn dictates that $b_{top} \leq N_0$. Applying our bounds for the bottom parts, we can bound the remainder c by a maximum value C , defined as $C = |4^m - 3^k|N_0 + (3^k - 1)/2 + (4^m - 1)/3$.

This shows that any valid sum $x < M \cdot N_0$ is uniquely determined by choosing a base value $y \in (A + B) \cap [0, N_0]$ and a remainder $c \in [0, C]$. Therefore, the total number of elements in $A + B$ strictly less than $M \cdot N_0$ is bounded by the product of the number of choices for y and the number of choices for c . Dividing by the interval length $M \cdot N_0$ yields an upper bound on the density at this new scale: the density up to $M \cdot N_0$ is less than or equal to the density up to N_0 multiplied by the factor $(C + 1)/M$.

To force the density to drop, we require this multiplying factor to be strictly less than 1. Because the ratio of logarithms $\ln(4)/\ln(3)$ is irrational, Dirichlet's Approximation Theorem guarantees that we can find arbitrarily large integers k and m such that the ratio $4^m/3^k$ is arbitrarily close to 1. By choosing k and m carefully, the absolute difference $|4^m - 3^k|$ becomes extremely small relative to M . Consequently, for any given N_0 , we can select k and m large enough such that $(C + 1)/M \leq 0.99$.

We can now apply this bounding process iteratively. Starting from an arbitrary scale N_0 , we can find a larger scale $N_1 = M_1 N_0$ where the density of $A + B$ drops by a factor of at least 0.99. Taking N_1 as our new baseline, we find a still larger scale $N_2 = M_2 N_1$ where the density drops by another factor of

0.99. After r iterations, the density of the set up to the scale N_r is bounded by $(0.99)^r$. As $r \rightarrow \infty$, this bound approaches 0. This demonstrates the existence of a sequence of arbitrarily large scales where the density tends to 0, proving that the lower density of $A + B$ is 0. $\square$